MIL-I-6048B (AER)
1 JUNE 1956
SUPERSEDING
MIL-I-6048A (AER)
1 OCTOBER 1951

MILITARY SPECIFICATION

INTERCOMMUNICATION SETS, AN/AIC-5A and AN/AIC-5B

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope - This specification establishes the requirements for equipment to be used to provide the pilot, co-pilot, and radio operator with selective control of all the radio communication and navigation equipments aboard an aircraft, and to provide selective interphone communications and some radio facilities for all crew members. The equipment is primarily for operation in patrol and transport aircraft.

1.2 Classification - The intercommunication set covered by this specification shall be of two types and shall consist of the following units in the quantities listed, unless specified otherwise in the contract:

AN/AIC-5A INTERCOMMUNICATION SET

Item	Unit	Type Designation	Max Allow. Size (in.) W	H/L	D	Max Allow. Wt. (lb)	Quantity
1	Operating Assembly	OA-53A/AIC-5	11	9-1/8	24	42.5	1
2	Control Unit (Master)	C-380A/AIC-5	9-5/16	4-1/2	3-1/2	3.8	3
3	Control Unit (Station)	C-379B/AIC-5	3-1/2	5-3/4	3-3/8	2.0	8
	Mountings						
4	Mounting Base	MT-778/AIC-5A	11-1/8	2-1/2	22	1.5	1

AN/AIC-5B INTERCOMMUNICATION SET

Item	Unit	Type Designation	Max Allow. Size (in.) W	H/L	D	Max Allow. Wt. (lb)	Quantity
1	Operating Assembly	OA-53A/AIC-5	11	9-1/8	24	42.5	1
2	Interphone Control (Master)	C-762A/AIC-5B	5-3/4	7-1/2	6-1/2	4	3
3	Interphone Control (Station)	C-761A/AIC-5B	5-3/4	2-5/8	6	1.5	8
	Mountings						
4	Mounting Base	MT-778/AIC-5A	11-1/8	2-1/2	22	1.5	1

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APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for ds, form a part of this specification to the extent specified herein:

2.2 Specifications:

Military

MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-E-5556	Enamel; Camouflage, Quick Drying
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-172	Cases and Mounting Bases, Electronic Aircraft
MIL-E-17555	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing, and Marking of
MIL-S-3950	Switches, Toggle (For Electronic and Communications Use)
JAN-T-152	Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic and Electrical Equipment; General Process for
MIL-F-25173	Fasteners; Control Panel, Aircraft Equipment

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EI-207	Control Panels, Electrical and Electronic Equipment, Rack or Console Mounted, Installation of
EI-247	Intercommunication Set AN/AIC-5B, Installation of
ET-247	Intercommunication Sets AN/AIC-5A and AN/AIC-5B Bench, Preflight, and Flight Test of

2.3 Drawings:

Air Force-Navy Aeronautical

MS 33558	Numerals and Letters - Aircraft Instrument Dial, Standard Form of
AN3219	Knob - Pointer, Small

Bureau of Aeronautics

EI-247-5 (Sheet 1 of 2)	OA-53A/AIC-5 Operating Assembly, Outline
EI-247-5 (Sheet 2 of 2)	OA-53A/AIC-5 Operating Assembly, Schematic

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54A1D55-8	C-380A/AIC-5 Control Unit, Outline and Schematic
EI-247-4 (Sheet 1 of 2)	C-379B/AIC-5 Control Unit, Outline
EI-247-4 (Sheet 2 of 2)	C-379B/AIC-5 Control Unit, Schematic
54A1D55-7	AM-171/AIC-5 Amplifier, (Interphone) Schematic
54A1D55-6	AM-172/AIC-5 Amplifier, (Isolation) Schematic
EI-247-3 (Sheet 1 of 2)	C-762A/AIC-5B Interphone Control (Master) Outline
EI-247-3 (Sheet 2 of 2)	C-762A/AIC-5B Interphone Control (Master) Schematic
EI-247-2	C-761A/AIC-5B Interphone Control (Station) Outline and Schematic

2.4 Availability of Documents:

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT.

2.5 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Bureau of Aeronautics.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of Parts and Materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern:

(1) Parts and Materials approved to specifications listed in MIL-E-5400 shall be given first consideration.

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(2) If the contractor can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the performance requirements of the respective standard specification, provided that no deterioration of size, weight, performance or reliability will result in the use of such substitutions.

(3) In all cases, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 must be of prime consideration.

(4) In the selection of parts for plug-in construction, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall first be approved by the Bureau of Aeronautics.

3.1.1 Non-Standard Part Approval and Material Approval - All non-standard parts or ıaterials (parts or materials which do not comply with MIL-E-5400) of an electrical nature or methods manufacture therefor and techniques and procedures involving their use and application shall be subıitted to the Bureau of Aeronautics for approval as required by Specification MIL-E-5400.

3.1.2 Workmanship - Workmanship used in the construction of this equipment shall be accordance with specification MIL-E-5400.

3.2 General Design Requirements:

3.2.1 General - This equipment shall meet all the general design requirements as specıed in Specification MIL-E-5400.

3.2.2 Installed Weight - The installed weight of each unit of the equipment shall be not ·eater than that listed in paragraph 1.2. The installed weight shall include cable plugs and clamps and .e mounting base, but shall not include microphone, headset, primary power supply, interconnecting ıbles, or mounting brackets.

3.2.3 Stability - The equipment shall operate with optimum performance for 50 hours, ɔntinuously or intermittently, without necessity of readjustment of any controls which are nonaccesble during flight.

3.2.4 Operating Life

3.2.4.1 Reliable Operating Life - The equipment shall have a reliable operating life of at ast 500 hours without requiring removal for bench servicing. Parts requiring servicing or replaceent at the end of this interval to renew this service life should be specified by the manufacturer.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life 2,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement withthis interval and the life of such parts shall be specified by the manufacturer.

3.2.5 Interconnecting Cabling - This equipment shall be capable of satisfactory operaɔn using unshielded interconnecting cabling. A small percentage of the individual wires may be ıielded, if necessary, provided the assembly of the cable to its plugs may be easily accomplished. easonable precautions (such as the inclusion of a filter in the equipment) shall be taken to prevent the

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unshielded wires of the interconnecting cables from producing fields of sufficient intensity to interfere with the satisfactory performance of the other electronic equipments normally installed in aircraft.

3.2.6 Cables and Connectors - Cables and connectors shall be in accordance with the requirements of Specification MIL-E-5400. Completed interconnecting cables are not required under this specification. All necessary connectors, cable clamps, ferrules, adapters, or other fittings required for fabrication of all cables shall be specified but not supplied by the contractor. Wire for cable fabrication shall not be supplied; however, cable makeup specifying size wire, necessary shielding, and other pertinent data shall be furnished. All connector inserts shall be melamine, or equivalent.

3.2.7 Console Controls - AN/AIC-5B console controls shall be in accordance with Specification MIL-C-6781. The brilliance of the lamp will be controlled externally and this control shall not be part of the control unit.

3.2.8 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions specified in Specification MIL-E-5400.

3.2.9 Power Requirements - The equipment shall not require power for operation other than from the following source, with characteristics as defined in Specification MIL-E-5400 with the exceptions as noted herein, nor require power in excess of the amount indicated.

DC Source — 28 Volts, 5 Amperes

3.2.9.1 Voltage Limits - The equipment shall be designed to meet the performance requirements specified herein for the following voltage limits:

3.2.9.1.1 DC Limits - The voltage limits shall be 24 to 29 volts for specified performance; however, the equipment must also operate from 17 to 24 volts with reduced performance permitted.

3.2.9.2 Grounded Input - The design of electronic equipment shall be such that the negative of the d-c power source may be grounded externally.

3.2.9.3 Power Source Transients - The equipment shall not be damaged when subjected to the applicable power source transients, and shall automatically resume normal operation at the expiration of the transient condition.

3.2.10 Overload Protection - Overload protection for the equipment shall be provided within the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum. Fuses shall be located in readily accessible places and shall be in accordance with Specification MIL-E-5400.

3.2.11 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that over-all spraying of the equipment is not necessary. Over-all spraying of the equipment in accordance with JAN-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.12 Panel Lettering - Lettering on AN/AIC-5A panels shall be white, and non-reflecting. It shall consist of enamel conforming to Specification MIL-E-5556 or shall provide delineation of proven value used on control boxes of conventional design. The wording shall be clearly and concisely indicative of function. The type characters shall conform to the numerals and letters as shown by Drawing MS 33558, or may be Compressed Commercial Gothic. The height of the characters shall be determined by the purpose, according to the following plan:

Purpose	Height (in.)
Group of controls or fuses	1/4
Identification of control or fuse	3/16
Positions of control - Jack markings	1/8

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ettering and markings on AN/AIC-5B panels shall conform to the requirements of Specification IL-C-6781.

3.2.13 Identification of Product

3.2.13.1 Nameplates - Nameplates and nomenclature shall be in accordance with Specification MIL-N-18307.

3.2.14 Interchangeability - The interchangeability requirements as specified in Specifiıtion MIL-E-5400 shall be extended to require interchangeability of units and component parts of this ıuipment with the latest model of any previously produced equipments. Such interchangeability shall ɜ measured against a model, manufacturing drawings, or other technical information as provided by :e Procuring Service for that purpose. Should the interchangeability requirements conflict with any ɜquirements of the general specification, the interchangeability requirements shall govern.

3.2.15 Provisions for Maintenance and Field Testing - Provisions for maintenance and ɜld testing shall meet the requirements of Specification MIL-E-5400.

3.2.16 Performance - The intercommunication sets shall provide the following basic ɔmmunication and operating features for aircraft crew members:

3.2.16.1 For the Pilot

(a) Permit selection of any one or combination of eight radio receiver outputs without individual readjustment of the control unit volume control.

(b) Permit selection of two navigation receiver outputs which shall be provided with range filter units under the control of a range filter switch.

(c) Permit selection of the microphone and control circuit of any three radio transmitters.

(d) Permit selective interphone communication to the headphone of either all crew stations or to the flight crew alone. During interphone communication, the output volume from any of the selected radio receivers shall be automatically reduced with respect to the interphone volume. Only the radio receiver outputs selected at each individual control unit shall be present as radio background during interphone communication.

(e) Permit limited range volume control which shall provide independent control of the radio communication receiver and interphone outputs.

(f) Permit immediate transfer of the headphones from normal to alternate connection by a simple switch operation.

(g) Permit control over the use of the VHF transmitter in AN/AIC-5A and UHF transmitter in AN/AIC-5B at flight and gun crew station controls.

(h) Provide microphone and headphone connection to the control unit directly, or through an external jack box (not supplied), or extension cord assembly (not supplied), as desired.

(i) Provide reception of tail warning device (TWD) signals insofar as possible without detriment to other functions.

(j) Provide for disconnecting the headphone from the ICS circuits to permit uninterrupted radio reception.

(k) Provide for overriding the radio operator ICS disconnect control [contained in (d) switch].

3.2.16.2 For the Co-pilot - Provide identical operating features as described for the ilot in paragraphs 3.2.16.1(a) to (i), inclusive, and in paragraph 3.2.16.1(k). Operating feature

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3.2.16.1(j) is not provided, and feature 3.2.16.1(k) enables overriding both the pilot and radio operator ICS disconnect control.

3.2.16.3 For the Radio Operator - Provide identical operating features as described for the pilot in paragraphs 3.2.16.1(a) to (j), inclusive. Operating feature 3.2.16.1(k) is not provided.

3.2.16.4 For Crew Members Other Than the Pilot, Co-pilot, and Radio Operator.

(a) Permit reception from VHF or UHF radio receiver output under control of a switch which allows disconnecting the radio receiver circuits at each control unit as desired.

(b) Permit selection of the microphone and control circuit of the VHF or UHF radio transmitter when the control circuit is completed by the pilot, co-pilot, or radio operator.

(c) Permit selective interphone communication for both flight crew and gun crew members which allows communication between members of the respective crews or with all crew members.

(d) Provide a limited range volume control which shall control both radio volume and interphone volume.

(e) Provide microphone and headphone connections to the control unit directly or through an external jack box (not supplied) or microphone-headset extension cord assembly (not supplied) and cabling to the control unit jack. A mask or boom microphone extension cord and jack shall be attached to each control unit. When the microphone-headset extension cord is utilized, short-circuiting of terminals inside the box is not permitted.

(f) Provide reception of TWD signals insofar as possible without detriment to other functions.

(g) Permit reception from one auxiliary radio receiver output. When utilized, this circuit shall be interrupted and the headphones shall be connected to interphone, if "ICS ALL" or "ICS EMG" is selected at any control or station box.

3.2.17 Components - Miniature and subminiature components shall be used where practicable.

3.2.18 Controls - Finger Clearance - The component parts shall be arranged on panels to give maximum finger clearances for the manipulation of controls.

3.2.19 Knob-Pointer - On AN/AIC-5A panels, selector switch knob-pointers shall be in accordance with Drawing AN3219, with control shafts flattened for their use.

3.2.20 Switch Position Cover Stops - If required, stops shall be provided for specified switch positions. Such stops for rotary selector type switches shall consist of a plate held in place by two small machine screws or other suitable device, covering the designated switch position, acting as a stop for the knob with which it is used, preventing operation of the switch to the position covered until the plate or device is removed, and not interfering in any manner with the operation of the switch to the other positions not covered. All sharp edges or projections that might cut the operator's hand or glove shall be removed from the cover plates and mounting screws. Provisions shall be made within each unit for stowing all switch position cover stops when they are not in use on the front panel.

3.2.21 Front Panel - The front panel of the control units shall be so designed as to prevent water getting into the unit through the panel if it should be subjected to direct incident rain even when the cover plates of paragraph 3.2.20 above are removed. All toggle switches shall be in accordance with MIL-S-3950 and shall be waterproofed by lever seals instead of external boots.

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3.2.22 Moisture Condensation Drainage

(a) Control units: One hole 1/8-inch diameter shall be drilled in each of the four corners of the side on which AN receptacles are mounted on the control unit.

(b) Operating Assembly: One hole 1/8-inch diameter shall be drilled in each of the four corners, and one hole 1/8-inch diameter shall be drilled approximately at the center of each side of the longest dimension on the bottom of the unit.

3.2.23 Relays - Relays shall be in accordance with the requirements of Specification IL-E-5400 and the following:

(a) When connected to 31 volts dc for a period of 4 hours in an ambient temperature of 60°C, the temperature rise in the armature winding shall not exceed the accepted safe limits for the type of insulating materials used.

(b) When in the nonoperated condition, all normally closed contacts shall remain closed and there shall be no noticeable motion of the relay armature when viewed with a standard stroboscope while the relay is vibrated in any direction at amplitudes as great as 0.03 inch (0.06 total excursion) at any frequency between 10 and 55 cycles per second.

(c) When operated on 17 volts dc, the operated contact shall remain closed and the armature shall remain firmly sealed to the armature stop, under the conditions of vibrations described in paragraph (b) above.

3.2.24 Frequency Characteristics

3.2.24.1 Isolation (or Receiving) Amplifiers - The frequency response of each isolation mplifier shall be within +1 db and -3 db of the 1,000 cps response at all frequencies between 300 and 000 cps. Frequencies below 300 cps and above 7,000 cps shall be attenuated a minimum of 3 db per ctave. The input voltage shall be 11 volts across the input network and the volume control shall be djusted to give 11 volts at 1,000 cps across a 600 ohm load, giving the required 0 db gain through the ystem.

3.2.24.2 Interphone Amplifiers - The frequency response of each interphone amplifier hall be within +1 db and -4 db of the 1,000 cps response at all frequencies between 300 and 4,000 cps. requencies below 300 cps and above 4,000 cps shall be attenuated a minimum of 4 db per octave. The nput shall be one volt and the volume control shall be adjusted to give 11 volts across the normal load 100 ohms.

3.2.25 Power Output

3.2.25.1 Isolation Amplifier - Each isolation amplifier shall deliver 200 milliwatts audio ower at 1,000 cps into a 600 ohm output termination with 11 volts across the input network.

3.2.25.2 Interphone Amplifier - Each interphone amplifier shall deliver 2 watts audio ower at 1,000 cps into a 60 ohm output termination with an input of 1 volt.

3.2.26 Impedances

3.2.26.1 Input - Amplifiers shall have input impedances as follows:

(a) Isolation Amplifier - 600 ohms

(b) Interphone Amplifier - Microphone - 3,300 ohms for connection to 180 ohm termination resistance.

(c) Radio - 1,000 ohms, nominal

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3.2.26.2 Output - Amplifiers shall have output impedances as follows:

(a) Isolation Amplifier - 600 ohms

(b) Interphone Amplifier - 60 ohms nominal; shall be designed to operate into any load between 50 and 300 ohms.

3.2.27 Distortion

3.2.27.1 Isolation Amplifier - The distortion in each isolation amplifier at 1,000 cps under the conditions described in paragraph 3.2.25.1 shall not exceed 5 percent.

3.2.27.2 Interphone Amplifier

(a) Normal operation - In normal operation, the distortion shall not exceed 9 percent at 1,000 cps with an input of 1 volt and an output of 11 volts across any load between 100 and 300 ohms.

(b) Alternate (emergency) operation - In alternate operation, the distortion at 1,000 cps with an input of 1 volt and an output of 11 volts across a load of 60 ohms shall not exceed 25 percent, provided all contributing distortion other than clipping shall not exceed 8 percent.

3.2.28 Gain Characteristics

3.2.28.1 Isolation Amplifier and Associated Isolating Network - Each isolation amplifier and associated isolating network shall be designed to deliver a net voltage gain of not less than 0 db from the output of the associated radio receiver to the output of the receiving amplifier terminated in 600 ohms under any or all of the following conditions:

(a) With the input voltage from the radio receiver output measured across the terminating potentiometer and the moving arm operated for maximum output.

(b) The resistance of the terminating potentiometer and associated limiting resistor at least 10 percent below the specified value.

(c) The resistance of the isolating (series) resistor at least 10 percent above the specified value.

(d) The resistance of the semivariable gain control of the receiving amplifier at least 10 percent below the specified value with the moving arm operated for maximum output.

(e) The mutual conductance of the receiving amplifier vacuum tube not exceeding 3,200 micromhos.

3.2.28.2 Interphone Amplifier

3.2.28.2.1 Input Voltage Variation - The overall output voltage variation shall not exceed ±1.5 volt for microphone input variation from 0.3 to 2.0 volts rms, and shall not exceed ±2 volts for radio input variation from 0.05 to 0.5 volts rms. The reference level shall be an input of 1 volt at 1,000 cps with the volume control adjusted to give an output of 11 volts across a 100 ohm load.

3.2.28.2.2 Output Impedance Variation - In normal operation, the output voltage shall not vary more than 1 volt with output impedances from 300 to 100 ohms. In emergency operation, the output voltage shall not vary more than 2.0 volts with output impedances from 300 to 50 ohms. Reference levels and adjustments shall be as indicated in paragraph 3.2.28.2.1.

3.2.28.3 Preadjustable Range - The range of preadjustable output voltage provided by a variable control shall be 7.8 to 11 volts rms (100 to 200 milliwatts per pair of headphones when the control unit or station box interphone volume control is adjusted for maximum output). This voltage shall be measured across a 100-ohm terminating impedance, with an input of 1 volt at 1,000 cps.

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3.2.29 Isolating Circuits - The isolation between any audio circuits not directly interıected (except through isolating resistors in the control boxes) shall be at least 40 db.

3.2.30 Radio Level During ICS - During interphone reception at any control unit, the ıut volume from any radio receivers selected at that control unit shall be automatically reduced 10 lb with respect to the interphone level.

3.2.31 Range Filter

3.2.31.1 "VOICE" Response - The insertion loss of the range filter in the "VOICE" ition shall not exceed 5 db, except that the response at 1,020 cps shall be 40 db below the response ,000 cps, and the response at 850 cps and 1,250 cps shall not be more than 10 db below the 2,000 response.

3.2.31.2 "BOTH" Response - In the "BOTH" position, the filter networks shall be :trically isolated from the communications circuits. The insertion loss in the "BOTH" position ll not exceed 0 db.

3.2.31.3 "RANGE" Response - The maximum allowable attenuation at 1,020 cps in the ANGE" position shall not be more than 5 db below the response at 1,020 cps with the filter in the)TH" position. The response at all frequencies below 850 cps and above 1,250 cps shall be more ı 6 db below the filter response at 1,020 cps. The response at frequencies below 600 cps and above)0 cps shall be attenuated more than 30 db down from the 1,020 cps response.

3.2.31.4 Accuracy of Frequencies - The frequencies specified in 3.2.31 shall be accurate vithin ±5 cps.

3.3 Detail Design Requirements:

3.3.1 Operating Assembly OA-53A/AIC-5 - The operating assembly unit shall fulfill the owing requirements:

3.3.1.1 Function - The OA-53A/AIC-5 operating assembly shall contain all the units l circuits necessary to the system other than switching circuits included in the master control units l station boxes. The functions of the various controls on the panel of the operating assembly are as lows:

3.3.1.1.1 "INCOMING RADIO LEVEL" Controls - These five controls designated "INMING RADIO LEVEL" as a group and individually as "COMM. No. 1", "COMM. No. 2", "COMM. . 3", "COMM. No. 4", and "U.H.F. No. 1" control the average output volume from the radio re-.vers which terminate on the switches similarly designated on the three master control units. The rpose of these controls is to permit adjusting the average output volumes of these five receivers to a :mmon volume level, and, thereby, to permit satisfactory mixing of various receiver outputs at the .ster control units without undesirable overloading effects from the receivers having unusually high :put volumes.

3.3.1.2 Emergency Switching Operation Controls

3.3.1.2.1 "ICS AUDIO OUTPUT" - These intercommunication sets include two interone amplifiers; one each for the flight-crew and gun-crew interphone circuits. During normal operons the "ICS AUDIO OUT" switches remain operated to "NORMAL", in which case each interphone ıplifier operates into its assigned interphone circuit at all times, "ICS ALL" and "ICS EMERGENCY" lls being accomplished by connecting the amplifier inputs together. In case of failure of the flight-ew interphone amplifier the "FLT. CR." switch is operated to "ALTERNATE", in which case all ght-crew headphones are removed from the flight-crew amplifier and connected to the gun-crew in-:phone circuit. Under this emergency condition the "crew" and "all" control lines shall connect gether resulting in "ICS ALL" operating in both "ICS CREW" and "ICS ALL" selector switch posi-)ns. Likewise, should the gun-crew interphone amplifier fail during flight, the gun-crew headphones ay be connected to the flight-crew interphone circuit by operating the "GUN CR." switch to ALTERNATE".

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3.3.1.2.2 "ISOLATION AMPLIFIER POWER" - These intercommunication systems include two dynamotor units for high-voltage supply. During normal operations both "ISOL. AMP. PWR." switches are operated to "NORMAL", in which case the pilot's isolation amplifier unit operates from one dynamotor and the co-pilot's and radio operator's isolation amplifiers operate from the second amplifier unit. Should one dynamotor fail during flight, the isolation amplifier or amplifiers supplied by that dynamotor can be connected to the remaining dynamotor unit by operating the "ISOL. AMP. PWR." switch associated with the control unit at which the failure was noted to "ALTERNATE". This operation shall also transfer the load from the associated interphone amplifier to the output of the other interphone amplifier, with operation as described in paragraph 3.3.1.2.1.

3.3.1.3 Isolation Amplifier AM-172/AIC-5 - This unit mounts in the operating assembly, one being required for each master control unit used. It prevents the mixed signals at each master control from reacting on the system, and compensates for losses in impedance matching.

3.3.1.4 Interphone Amplifier AM-171/AIC-5 - This two-stage audio amplifier provides the necessary automatic gain characteristics for feeding interphone signals originating at the microphone of any station to all stations. It shall operate over a wide range of input voltages and output impedances.

3.3.1.5 Form Factor and Mounting - The operating assembly unit shall be assembled into case size B1-D1 (in accordance with Specification MIL-C-172) and shall be vibration mounted. Dimensions and arrangements shall be in accordance with Drawing EI-247-5 (sheet 1 of 2).

3.3.1.6 Contents of Unit - The operating assembly shall contain the following:

Quantity	Part
2	AM-171/AIC-5 Interphone Amplifier Unit, individually removable for maintenance.
3	AM-172/AIC-5 Isolation Amplifier Unit, individually removable for maintenance.
2	DY-2/ARR-2 Series Dynamotor.
9 + 1 Spare	Hermetically sealed, plug-in type, DPDT Relay.
9 + 3 Spares	Screw Type 28-volt panel fuse.
6 + 3 Spares	Screw Type 250-volt panel fuse.
1	Receptacle labeled "ACCESS." to supply 28 V and 250 V to accessory (including a receptacle cap).

3.3.1.7 Weight - The total weight of the operating assembly unit, including vibration mounts, shall not exceed that listed in paragraph 1.2.

3.3.1.8 External Electrical Connections - Electrical connections to the amplifier unit shall enter through one AN3108-20-29S, one AN3108-20-29S-W, and one AN3108-20-29S-Z connector, each fitted with an AN3057-12 cable clamp. The connectors shall be differently keyed to prevent improper connections. As a further aid in correctly mating the plugs and receptacles, they shall be coated externally with Specification MIL-E-5556 enamel as follows: Left-hand plug and receptacle looking at the face of panel (P-101 and J-101)-red; center plug and receptacle - blue; and right-hand plug and receptacle (P-103 and J-103)-yellow. The threaded portions of the connectors shall not be painted. In addition, a receptacle shall be mounted on the front panel of the amplifier unit to provide connections for the LM frequency meter for testing purposes, or other accessories.

3.3.1.9 Schematic Design - The operating assembly unit shall be designed in accordance with Drawings EI-247-5 (2 sheets), 54A1B55-6 and 54A1B55-7.

3.3.1.10 Controls and Adjustments - The following controls and adjustments labeled as capitalized below, shall be provided on the front panel of the operating assembly:

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3.3.1.10.1 "INCOMING RADIO LEVEL" - This group of five knob controls individually sets the radio-communication-receiver audio output voltages from "COMM. No. 1", "COMM. No. 2", "COMM. No. 3", "COMM. No. 4", and "U.H.F. No. 1" receivers.

3.3.1.10.2 "ICS AUDIO OUT" - Two separate two-position toggle switches "FLT. CR." and "GUN CR." In the upper "NORMAL" position the flight crew and gun crew are connected to separate interphone amplifiers, while in the lower "ALTERNATE" position the flight crew is connected to the gun-crew interphone amplifier and the gun crew is connected to the flight-crew interphone amplifier.

3.3.1.10.3 "ISOL. AMP. PWR." - Two separate two-position toggle switches "PILOT" and "CO-P. and R. OPR." In the upper "NORMAL" position the isolation amplifiers are connected to separate dynamotors while in the lower "ALTERNATE" position the isolation amplifiers are connected to opposite dynamotors.

3.3.2 Master Control Unit C-380A/AIC-5 - Each master control unit shall fulfill the following requirements:

3.3.2.1 Function - Master control units shall be provided for the use of the pilot, co-pilot and radio operator. These units shall permit selecting any one or combination of eight radio receiver outputs as desired, selection of one of two navigation receivers each having a range filter, transmission over any one of three radio transmitters, and interphone communication to other members of the flight crew only (pilot, co-pilot, radio operator, and any additional crew members as selected in the original layout of the aircraft system, such as the navigator, radar operator, etc.) or to all crew members aboard. Combined with these general features are special provisions designed to expedite operation under combat conditions which will be described in detail later in this section.

3.3.2.1.1 Headphone Circuit Selector Switch - A headphone selector switch (designated "PHONES") shall be located in the upper left-hand corner of each master control unit. When this switch is operated to "NORMAL", the headphone circuit shall connect to the various receiver output and interphone circuits selected at the associated master control unit. When the "PHONES" switch is operated to "ALTERNATE" at the pilot's position, the pilot's headphone shall be transferred to the receiver and interphone circuits as selected at the co-pilot's master control unit. Similarly, when the "PHONES" switch is operated to "ALTERNATE" at the co-pilot's position, his headphones shall be transferred to the radio operator's control unit; and when the "PHONES" switch is operated to "ALTERNATE" at the radio operator's position, his headphones shall be transferred to the pilot's control unit.

3.3.2.1.2 "RECEIVER OUTPUT SELECTION" Switches - Any one or combination of eight radio outputs shall be selected by operating the toggle switch associated with each receiver (as indicated by the designations across the top of the master control unit) up toward the designations, which are the "ON" positions.

3.3.2.1.3 Range Filter Selector Switch - The output from each navigation receiver may or may not be operated through the "range" or "voice" sections of a range filter, depending upon the position of the range filter selector switch. Since only one navigation receiver at a time is used by any one crew member to ride the radio range, selection for both navigation receivers shall be combined into a single selector switch "RANGE FILTER SEL.", which operates as follows:

Circuit Arrangements for Various Positions of Range Filter Selector Switch

Resulting Connections of the Range Filter Unit Associated With Navigation Receivers

When "RANGE FILTER SEL." Is Operated to	Navigation 1	Navigation 2
"1-V"	Voice only received	Range Filter not connected
"1-R"	Range Signal only received	Range Filter not connected

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When "RANGE FILTER SEL." Is Operated to	Navigation 1	Navigation 2
"1-BOTH"	Both voice and range signal received	Range Filter not connected
"OFF"	Receiver not connected to Phones	Receiver not connected to Phones
"2-BOTH"	Range Filter not connected	Both voice and range signals received
"2-R"	Range Filter not connected	Range Signal only received
"2-V"	Range Filter not connected	Voice only received

Thus, when the "RANGE FILTER SEL." switch is operated to "2-R" position, the output from Navigation Receiver No. 1 shall be disconnected from the headphone circuit and the "range" section of the range filter associated with Navigation Receiver No. 2 shall be inserted between the output of Navigation Receiver No. 2 and the headphone circuit, thus permitting only the 1,020 cycle range tone to be transmitted to the headphone and suppressing voice signals. Likewise, when the range switch is in the "1-V" or "2-V" position, the 1,020-cycle reject filter section shall be connected into the navigation receiver output circuit selected, thus causing the 1,020-cycle range signal to be suppressed and the voice signal passed. When the range switch is in the "1-BOTH" or "2-BOTH" position, or the range filter is shown as not connected in the above table, the range filter shall be completely disconnected from the navigation receiver outputs or input respectively, and, therefore, shall not affect the performance of the navigation receivers in any way.

3.3.2.1.4 Incoming Interphone Volume - The volume level of incoming interphone calls shall be adjusted by means of the interphone volume control ("ICS VOL.") on the master control unit. It will usually be necessary to adjust this control only to allow for varying ambient acoustic noise levels, due to the design of the interphone amplifier which automatically levels varying input volumes due to differences in the speaker's voice, microphone efficiencies, altitude effects, etc., to a constant output volume.

3.3.2.1.5 Radio Volume Control - The radio volume control ("RADIO VOL") shall adjust the volume level in the headphones of any communication or VHF radio receiver outputs which are selected. When the navigation or marker beacon receiver outputs are selected, these outputs shall be connected directly to the headphone circuit and the volume received shall not be adjusted by means of the radio volume control on the master control unit. For this reason, all control of the output volume from either navigation receiver must be made by adjustment of the sensitivity control located on the tuning unit associated with each navigation receiver.

3.3.2.1.6 ICS Normal-Disconnect Switch - This switch which is made operative for the pilot and radio operator only, provides normal reception in the "NORMAL" position, and in the "DISCONNECT" position excludes the master control unit from all but emergency interphone communication. Under this condition, nonemergency interphone communication will not be received and interphone calls cannot be initiated until the switch is returned to "NORMAL" position. This switch shall be safety-wired to "NORMAL" at the co-pilot's position. The initiation of interphone calls to override the disconnect feature, when in use, is described below.

3.3.2.1.7 Interphone Circuit Selection Switch - Three positions of the six-position "ICS AND MIC. SEL." switch shall be utilized for interphone connections. Operation at these three positions shall be as follows:

3.3.2.1.7.1 "ICS CREW" Calls - Since the majority of interphone calls from master control unit stations involve the other master control unit stations (pilot, co-pilot, and radio operator) and a limited number of additional stations involved in the flight status of the aircraft (navigator, radar operator, countermeasures operator, etc.), these stations shall be grouped on a single interphone circuit known generally as the "flight crew" interphone but designated on the master control unit for brevity as "ICS CREW". Therefore, when it is desired to initiate an interphone call where members

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of the flight crew only are involved, the "ICS AND MIC. SEL." switch shall be operated to "ICS CREW" position. Under this condition, when the "push-to-talk" button of the associated microphone is operated, the flight crew interphone system shall be energized and the call shall be heard by all members of the flight crew but will not override the disconnect feature on the pilot's and radio operator's control units, or the "RCVR" position of the station control unit "MIC. SEL." switch.

3.3.2.1.7.2 "ICS ALL" Calls - When it is desired to initiate interphone communication to all crew members aboard, the "ICS AND MIC. SEL." switch shall be operated to "ICS ALL" position. Under this condition, when the "push-to-talk" button of the associated microphone is operated, the entire ship's interphone system shall be energized and the call shall be heard by all crew members aboard, but shall not override the disconnect feature on the pilot's and radio operator's control units.

3.3.2.1.7.3 "ICS EMG" Calls - When either the pilot or co-pilot (particularly the co-pilot when the pilot is in the interphone disconnect condition) desires to initiate interphone communication of such urgency that it is essential all crew members aboard be cognizant of the message regardless of any individual adjustments of the interphone system, the "ICS AND MIC. SEL." switch shall be operated to the "ICS EMG" position. Under this condition, when the "push-to-talk" button of the associated microphone is operated, the entire ship's interphone system shall be energized the same as for an "ICS ALL" call, but with the additional feature that the pilot's and radio operator's disconnect feature will be overridden and these crew members added to the interphone circuit for the duration of call regardless of the position of their respective "ICS AND MIC. SEL." switches. Under these conditions, the pilot or radio operator disconnect feature is overridden only for the duration of the interphone call from the station initiating the emergency call. Immediately upon release of the interphone circuit by the station making the emergency call, any answering call will not be heard at the pilot's or radio operator's position unless the "ICS NORMAL-DISCONNECT" switch is operated away from the "DISCONNECT" position, with one exception that the radio operator may hear the pilot answer an emergency call from the co-pilot provided the pilot has operated the "ICS AND MIC. SEL." switch to the emergency position when making the answering call. "ICS ORIDE" (override) shall be utilized in the AN/AIC-5B in lieu of "ICS EMG."

3.3.2.1.7.4 Omission of the Emergency Call Feature - In a standard system installation, the emergency interphone call feature will not be used at the radio operator's position since it is considered that the pilot's disconnect should be overridden only when the urgency of a particular message from some other crew member has been determined by the co-pilot to justify such a procedure. This shall be accomplished by covering the "ICS. EMG." position by a cover plate stop.

3.3.2.1.7.5 Radio Transmitter Selection - When the "ICS AND MIC. SEL." switch is operated to any one of the three radio transmitter positions, microphone and control circuits shall be connected directly to the desired transmitter as indicated by the designations associated with this switch. Under this condition, the selected radio transmitter can be energized and modulated by operating the "push-to-talk" button of the associated microphone unit and speaking into the microphone.

3.3.2.1.8 Crew Transmit Feature - A "CREW TRANSMIT-OUT" switch shall be provided on each master control unit to restrict the use of the UHF transmitter. When in the "OUT" position, only the pilot, co-pilot, and radio operator can use the UHF transmitter. In the "TRANSMIT" position, this transmitter can be used by other crew members.

3.3.2.1.9 Microphone and Headphone Connections - Microphone and headphone jacks shall be provided on each master control unit In most installations these jacks will be for emergency use only, and more convenient microphone and headset connections will be provided through a separate jack box or a headset-microphone cord.

3.3.2.2 Form Factor and Mounting - The over-all form factor, arrangement, and mounting shall conform to Drawing 54A1D55-8.

3.3.2.3 Schematic Design - The master control unit shall be designed in accordance with Drawing 54A1D55-8.

3.3.2.4 Controls and Adjustments - The following controls and adjustments labeled as capitalized below shall be provided on the front panel of the master control unit:

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(a) "PHONES" - A two-position toggle switch. In the upper "NORMAL" position phones are connected normally, while in the lower "ALTERNATE" position the phones are connected to the output of other master control units.

(b) Receiver audio switches - These eight switches shall be labeled left to right as follows: "BEACON" "1", "2", "3", "4", "U.H.F.", "SP. 1", "SP. 2" with "COMMUNICATION RCVRS." above the 1 to 4 designations. Switches shall be two-position toggle type with off in the lower position, and shall connect audio output from the listed receivers to the headphones. "-OUT-" shall be shown under these switches to indicate the off positions.

(c) "RANGE FILTER SEL." - This switch shall be a seven-position rotary type with "2V", "2R", and "2-BOTH" on left side, "OFF" in center, and "1-BOTH", "1R", and "1V" on right side in a clockwise direction.

(d) "ICS VOL." - A knob control for controlling interphone output to the headphones.

(e) "RADIO VOL." - A knob control for controlling radio level to headphones.

(f) "ICS" - A two-position toggle switch with "NORMAL" in the upper position and "DISCONNECT" in the lower position.

(g) "ICS AND MIC. SEL." - A six-position rotary switch with "ICS EMG", "ICS ALL", "ICS CREW", "U.H.F. TRANS.", "H.F. TRANS" and "SP." positions clockwise. A cover plate stop for the "EMG" position shall be provided for each master control unit.

(h) "CREW" - A two-position toggle switch with "TRANSMIT" in the upper position and "OFF" in the lower position.

3.3.2.5 Weight - The total weight of the master control unit and mounting base shall not exceed that listed in paragraph 1.2.

3.3.2.6 External Electrical Connectors - Electrical connections to the master control unit shall enter through one AN3106-20-29S and one AN3106-20-29S-W connector, each differently keyed to prevent improper connection, and fitted with an AN3057-12 cable clamp. As a further aid in correctly rectly mating the plugs and receptacles, they shall be coated externally, with Specification MIL-E-5556 enamel as follows: Left-hand plug and receptacle looking at face of unit (P-201 and J-201) - red, and right-hand plug and receptacle (P-202 and J-202) - blue.

3.3.3 Interphone Control C-762A/AIC-5B - Console control C-762A/AIC-5B shall conform to Specification MIL-C-6781, shall have fastening devices which conform with MIL-F-25173 and shall provide for mounting in accordance with specification EI-207. Console Control C-762A/AIC-5B shall fulfill the requirements for the C-380A/AIC-5 master control unit with the following exceptions and additions:

(a) Microphone and headset jacks (paragraph 3.3.2.1.9) are not required.

(b) Form factor and mounting (paragraph 3.3.2.2) does not apply. The form factor, mounting, wiring, lettering and panel arrangement shall be in accordance with Drawing EI-247-3 (2 sheets).

(c) An AN3106-22-14S connector shall be used in lieu of AN3106-20-29S-W.

3.3.4 Station Control Unit C-379B/AIC-5 - Each station control unit shall fulfill the following requirements:

3.3.4.1 Function - The C-379B/AIC-5 station control unit shall be provided for the use of all crew members aboard except the pilot, co-pilot, and radio operator. The following functions shall be provided:

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3.3.4.1.1 Volume Control - The single volume control "VOLUME" shall control the incoming volume from radio receivers (when selected) and the interphone circuit. It will usually be necessary to adjust the volume control only to allow for varying ambient acoustical noise levels since the UHF receiver output as well as interphone calls are fed to all station boxes through the interphone amplifier which automatically levels variations in input volume to a constant volume.

3.3.4.1.2 Receiver Output - The output from the UHF receiver shall be available at each station box during the interval between interphone calls. When the "MIC. SEL." switch is operated to the designation "UHF", the output from the UHF receiver shall be heard, except when interrupted by interphone calls. Operation of "MIC. SEL." switch to any ICS position shall remove the UHF radio receiver output from the headphone at that station box. Operation to "RCVR" shall connect a special radio receiver output to the headphones, which shall be interrupted only on "ICS ALL" or "ICS EMG" calls.

3.3.4.1.3 Radio Transmitter Selection - When interplane communication is desired, crew-members' station box positions may transmit over the UHF radio transmitter by operating the "MIC. SEL." switch to UHF position, operating the "push-to-talk" button of the associated microphone, and proceeding with radio communication, provided that such transmission is permitted by the operation of a switch on one of the master control units.

3.3.4.1.4 Interphone Circuit Selection - In order to provide more freedom of interphone communication between members of the gun crew and to avoid interference with radio reception by flight-crew members during such communications, all crew members not assigned to the flight-crew interphone system shall be grouped on a separate interphone system.

(a) "ICS CREW" calls - When it is desired to initiate an interphone call where members of the flight crew are not involved, the "MIC. SEL." switch shall be operated to "ICS CREW". Under this condition, when the associated microphone "push-to-talk" switch is operated, the interphone circuit involving gun crew members only is energized and the interphone call will be heard at gun-crew stations only.

(b) "ICS ALL" calls - When it is desired to initiate interphone communication to all crew members aboard, the "MIC. SEL." switch is operated to "ICS ALL" position. Under this condition, when the associated microphone "push-to-talk" button is operated, the entire ship's interphone system is energized and the call will be heard by all crew members aboard, but the disconnect feature on the pilot's and radio operator's control units shall not be overridden.

(c) "ICS EMG" calls - Since station boxes are not provided with an emergency interphone circuit, emergency calls shall be made using the procedure for "ICS ALL" calls described in paragraph 3.3.2.1.7.2. Such calls will be received by the co-pilot (who is not provided with an interphone disconnect circuit) even though both the pilot and radio operator may be disconnected from the interphone circuit.

3.3.4.1.5 Microphone and Headphone Connections - Microphone and headphone connections to each station box shall normally be made directly to the microphone (MIC) or headphone (PHONE) jacks on the station box, except that when mask or boom microphones are being used, the microphone connection shall be made to the JJ-055 type jack on the end of the microphone extension cord which projects from the station box between the microphone and headphone jacks. In those cases where a headset-microphone cord set is provided at a station box position, the microphone and headphone connections shall be normally made to this cord, and it will be noted that the microphone extension cord is usually removed from the station box. In either case, the microphone and headphone jacks on the station box shall be used for emergency connections when required.

3.3.4.2 Form Factor and Mounting - The overall form factor arrangement and mounting shall conform to Drawing EI-247-4 (sheet 1 of 2).

3.3.4.3 Schematic Design - The station control unit shall be designed in accordance with Drawing EI-247-4 (sheet 2 of 2).

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3.3.4.4 Controls and Adjustments - The following controls and adjustments labeled as capitalized below, shall be provided on the front panel of the station control unit:

(a) "VOLUME" - A knob control for setting interphone or radio level to the headphones.

(b) "MIC. SEL." - A four-position rotary switch connecting microphone and headphones to "ICS ALL", "ICS CREW", or "UHF", or headphones alone to "RCVR". The "RCVR" position shall be covered by a removable cover plate stop to prevent use of this postion when not required.

3.3.4.5 Weight - The total weight of the station control unit and mounting base shall not exceed that listed in paragraph 1.2.

3.3.4.6 External Electrical Connections - Electrical connections to the station control unit shall enter through an AN3106-20-29S connector fitted with an AN3057-12 cable clamp.

3.3.5 Interphone Control C-761A/AIC-5B - Console control C-761A/AIC-5B shall conform to Specification MIL-C-6781, shall contain fastening devices in accordance with MIL-F-25173, and shall be capable of mounting in accordance with Specification EI-207. Console Control C-761A/AIC-5B shall fulfill the requirements for the C-379B/AIC-5 station control unit with the following exceptions and additions.

(a) Mask or boom microphone extension cord and jack (paragraphs 3.2.16.4(e) and 3.3.4.1.5) are not required.

(b) Microphone and headset jacks (paragraph 3.3.4.1.5) are not required.

(c) Form factor and mounting (paragraph 3.3.4.2) does not apply. The form factor, mounting, wiring, lettering, and panel arrangement shall be in accordance with Drawing EI-247-2.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests

(1) Preproduction Tests - Preproduction Tests are those tests conducted on an equipment representative of the equipments to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction Tests shall be accomplished by Contractor Demonstration Tests and Service Approval Tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests shall be accomplished under the responsibility of the contractor to demonstrate that the equipment complies with all applicable requirements.

(b) Service Approval Tests - Service Approval Tests shall be accomplished under the responsibility of the procuring service for the purpose of verifying the results of the Contractors Demonstration Tests and for conducting any additional tests deemed necessary to determine that the equipment complies with all applicable requirements.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on equipment submitted for acceptance under the contract. Acceptance Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics and which have previously passed Individual Tests.

(c) Special Tests - Special Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics.

4.2 Preproduction Tests -

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more equipments as necessary to determine that the design as proposed by the contractor will meet all the requirements of the applicable specifications and the contract. Contractor Demonstration Tests shall be conducted in accordance with the approved test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the equipment for Service Approval Tests. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. When tests are supervised the Bureau of Aeronautics may, when conducting Service Approval Tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting Service Approval Tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary. Contractors not having laboratory facilities to conduct all required tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

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4.2.2 Service Approval Tests - The Service Approval Sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for Service Approval Tests, the contractor shall also submit the accessory material and design and test data specified in MIL-E-5400. The design and test data shall indicate the physical and electrical characteristics of the equipment and establish that the equipment meets all applicable requirements. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction Tests shall include environmental tests in accordance with MIL-T-5422 and interference tests in accordance with MIL-I-6181. The testing of plastic lighting plates shall be in accordance with MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance under the contract. The preproduction samples shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Bureau of Aeronautics shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipments as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests and has been packaged and packed ready for shipment.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as approved or requested by the Bureau of Aeronautics.

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4.3.1 Individual Test - Each equipment submitted for acceptance shall be subjected to the Individual Test to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted must successfully pass the tests specified in the following paragraph.

4.3.1.1 Scope of Test - As a minimum, each equipment shall be subjected to the following:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1.1 Examination of Product - Each equipment shall be carefully examined to determine that the requirements for material and workmanship as specified in MIL-E-5400 has been met. Also, it shall be determined that approval has been granted for the use of all non-standard parts.

4.3.1.1.2 Operational Test - Each equipment shall be operated. The period of operation shall be long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.1.3 Reliability Test - Each equipment shall be operated under the conditions specified below for a period of 12 hours without failure A failure shall be defined as anything which causes malfunctioning of the equipment. Only such adjustments shall be permitted which can be made using such controls and adjustments which would normally be accessible during flight.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (15 minute) (Without Isolators)	Any frequency within the range of 20 to 60 cps and an amplitude of ± 5 g's

The fifteen (15) minute vibration run shall be made at the beginning of the 12 hour period. The equipment shall be operating during the vibration run. During the twelve (12) hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur it shall be repaired and the test started over, except the fifteen (15) minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action must be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.3.2 Sampling Test - Sampling Test shall be conducted on a quantity of equipment as specified in the following paragraph. Equipment selected must first have passed all of the individual tests. Sampling Tests shall be conducted on complete equipments.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment. Selection of equipment for tests shall be under the supervision of the naval inspector.

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Quantity Delivered for Acceptance	Quantity to be tested
From the first 10	1
From 11 to 100	2
From 101 to 200	1
From 200 and above	1 for each additional 200 or fraction thereof

4.3.2.2. Scope of Test - As a minimum each equipment selected for Sampling Test shall be subject to the tests specified below.

Tests	
I	Conduct a complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure all applicable specification requirements have been met.
II	Repeat the above using the highest and lowest specified input voltages and frequencies.
III	Conduct the reliability test specified in paragraph 4.3.1.1.3 except the test duration shall be 50 hours and two failures shall be permitted.

4.3.3 Special Test - Special Tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. Equipment selected for Special Test may be selected from equipment previously subjected to the Sampling Test.

4.3.3.1 Special Test Schedule - In selecting equipment for Special Tests the following schedule shall be followed:

(1) An early production equipment.

(2) An equipment midway of the production run.

(3) An early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional tests are necessary.

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

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4.3.3.2 Scope of Test - In conducting Special Tests, test procedures previously approved for the Preproduction and Sampling Tests shall be used where applicable. When not applicable the contractor shall prepare a test procedure and submit to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum the following tests shall be made, except that for those equipment selected as a result of design changes and failure reports only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (90 minutes cycling - 30 min. in each of 3 planes (Freq 5 to 500 cps ± 5g)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the Sampling or Special Test, the following corrective action shall be taken.

(1) Determine the cause of failure

(2) Determine if the failure is an isolated case or design defect.

(3) If a design defect, suggest corrective action for Bureau of Aeronautics approval.

(4) Where practical include a test in the Individual Test to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting Preproduction Tests, Acceptance Tests and Life Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.5 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements. The design data submitted by the contractor for the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

4.6 Rejection and Retest - Electronic equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

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4. Equipment Changes - After approval of the preproduction model no changes shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This applies for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with the requirements of SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specifications MIL-E-17555 and MIL-STD-129.

6. NOTES

6.1 Test Values --Normal and limiting values of performance data shall be determined at input voltages of 27.5 volts dc. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standars of performance.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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